

Exposure and Residue Correspondence: Two Silent Failure Modes When Comparing AlphaFold Database Models with Experimental Structures

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Abstract

Comparisons of AlphaFold Database (AFDB) models with experimental structures underpin knowledge of pLDDT, AlphaFold’s prediction of the local distance difference test (IDDT). Two problems can distort them silently: exposure, when the reference or a close relative was available to the model in training or as a template, and wrong residue correspondence, when residues are paired with wrong reference atoms. Of 868 ATLAS proteins, 791 (91%) had no reference released after AlphaFold2’s training cutoff. In the first four batches of a preregistered census of X-ray PDB entries released after 2026-09-11, 87 of 138 proteins (63%) had a same-protein or $\geq 95\%$ -identical structure predating the cutoff, 10 (7%) had no relative at $\geq 30\%$ identity, and 45 models (33%) were given a same-protein template. In an earlier, exploratory census, proteins with such relatives had slightly higher IDDT and pLDDT and no detectable difference in confident errors (pLDDT ≥ 90 , IDDT < 0.60). Our pipeline showed the correspondence error: for some chains it mapped residues in label numbering but looked atoms up in author numbering. Correction changed the median IDDT of 188 of 488 chain segments (151 by more than 0.2) and removed spurious confident errors mimicking overconfidence. A preregistered coverage test of conformal intervals for IDDT, which does not use pLDDT, failed on original values; we withdraw that outcome. A rerun registered after the error and a preliminary result were known meets the preregistered rule on corrected values: mean coverage 0.8945 and 0.9456 at nominal 0.90 and 0.95, against thresholds of 0.8914 and 0.9449.

1 Introduction

AlphaFold2 [Jumper et al., 2021] and the AlphaFold Protein Structure Database (AFDB; Varadi et al., 2022, Bertoni et al., 2026) have made predicted structures available for hundreds of millions of proteins. For each residue, AlphaFold reports a confidence score that predicts the local distance difference test (IDDT), a superposition-free score of local agreement between a model and a reference structure [Mariani et al., 2013]. The score is called predicted IDDT (pLDDT), and much of what is known about it comes from comparing AFDB models with other data on the same protein. Comparisons with experimental structures, simulations and disorder annotations ask how well pLDDT tracks local error [Jumper et al., 2021], whether it reflects flexibility or disorder [Akdel et al., 2022, Ruff and Pappu, 2021, Guo et al., 2022, Saldaño et al., 2022, Carugo, 2023], and whether conformal intervals built on pLDDT reach their nominal coverage [Shihab et al., 2026]. Two problems can make such comparisons misleading without stopping the pipeline. Agreement can look too good because the model has seen the reference or something close to it. It can look too bad because model residues were paired with the wrong reference atoms.

The first problem is *exposure*: the reference or a close relative could have informed the model, as a training example, as a *template* (a known structure given to the network as input at prediction time), or as a close homologue in the training data. For some questions, such as describing AFDB as it is used, exposure need not invalidate the comparison. For others, such as accuracy on new proteins or the calibration of pLDDT on unseen folds, agreement with an exposed reference may reflect recall rather than generalisation. When the

claim is generalisation beyond exposed targets, this is a form of data leakage [Kaufman et al., 2012, Kapoor and Narayanan, 2023], and we call the routes by which it can enter an AFDB comparison *exposure channels*. Structure prediction has long guarded against leakage through blind and time-split evaluation [Kryshtafovych et al., 2021, Haas et al., 2018, Jumper et al., 2021], and recent benchmarks show that release-date splits alone still leak through homology [Buttenschoen et al., 2024, Durairaj et al., 2024, Dobson et al., 2025, Bushuiev et al., 2024]. In a retrospective AFDB comparison the model is fixed and the templates it was given are recorded in its model file, so template exposure can be measured rather than assumed.

The second problem is *residue correspondence*. Model residues are numbered by their position in the UniProt sequence. Reference residues are numbered within an entry of the Protein Data Bank (PDB), whose mmCIF (macromolecular Crystallographic Information File) format carries two schemes: a sequential *label numbering* (`label_seq_id`) and an *author numbering* (`auth_seq_id`) chosen by the depositors [Westbrook et al., 2022, RCSB Protein Data Bank, 2026a, wwPDB, 2026]. Resources such as SIFTS (Structure Integration with Function, Taxonomy and Sequences), distributed by the Protein Data Bank in Europe (PDBe), map these numberings to UniProt (Section 2). Less attention has gone to what happens downstream when a pipeline looks atoms up in one scheme using positions given in the other. Forensic re-analyses in high-throughput biology have found that the most common errors are simple ones, such as row or column offsets [Baggerly and Coombes, 2009]. A shifted correspondence produces numbers that look like results.

We measure the first problem in two corpora and report an instance of the second from our own pipeline. Our contributions are the following.

1. A decomposition of exposure into training eligibility, template exposure and homology exposure, with a measurement for each. For AFDB models the template list can be read from the model file (Section 3).
2. In ATLAS [Vander Meersche et al., 2024], a molecular-dynamics corpus that has been reused to train and evaluate structure-ensemble generators [Jing et al., 2024], 791 of 868 proteins have only references released on or before AlphaFold2’s training cutoff of 2018-04-30 (Section 5).
3. Before any of its entries had been released, we preregistered a census of X-ray PDB entries released after 2026-09-11, collected in weekly batches: its design, code and predictions were fixed and timestamped. In its first four batches, covering releases up to 2026-10-04, 87 of 138 proteins already had a structure of the same protein or a $\geq 95\%$ -identical one in the *training-era PDB* (entries released on or before the training cutoff), and 45 of the 138 AFDB models were given a same-protein template. In an earlier, exploratory census of entries released after 2022-06-02, proteins with such a relative had slightly higher IDDT, higher pLDDT by a larger margin and a share of *confident errors* (residues with $\text{pLDDT} \geq 90$ and $\text{IDDT} < 0.60$) that we could not distinguish from that of other proteins (Section 6).
4. In our census pipeline, a fallback that built the residue map in label numbering while looking atoms up by author numbering shifted the pairing of whole runs of residues and created residues with high pLDDT and low IDDT. A preregistered prediction about the coverage of conformal intervals for IDDT, which do not use pLDDT, failed on these values, and we withdraw that outcome as invalid. A rerun on corrected values, registered after the error was found and after a preliminary result was known, meets the registered decision rule at both nominal levels (0.90 and 0.95), and at 0.95 exceeds its threshold by only 0.0007. We give the mechanism, how often it is triggered and the checks that catch it (Section 7, Figure 1).
5. A table of practical checks (Section 8) and a public release of the registrations, code and derived tables.

Sections 2–4 give related work, the exposure channels and the methods, Sections 5–7 the results, and Sections 8–9 recommendations and limitations.

2 Related work

Leakage in machine learning for science. Leakage is a recurring cause of irreproducible results in machine-learning-based science [Kaufman et al., 2012, Kapoor and Narayanan, 2023]. Biology has its own guidance on validation and data splitting [Walsh et al., 2021, Whalen et al., 2022] and tools for splitting biomolecular data by similarity [Joeres et al., 2025]. In structural biology, standard train–test splits leak through sequence and structural similarity in protein–ligand affinity prediction [Li and Yang, 2017, Volkov

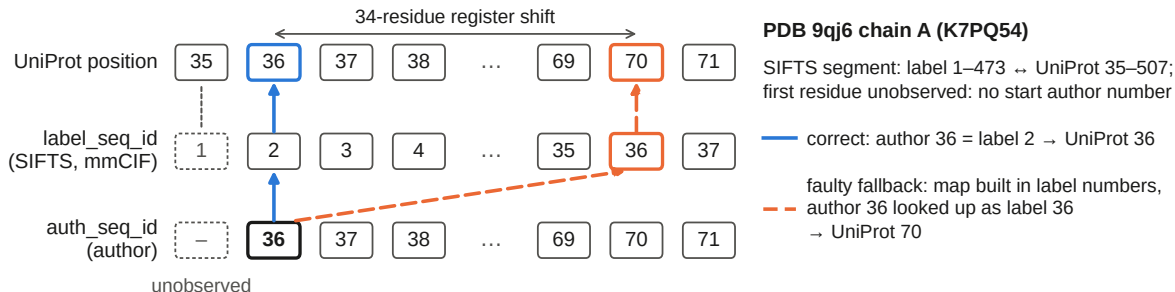


Figure 1: How the numbering fallback shifted PDB entry 9qj6, chain A (protein K7PQ54). Its SIFTS segment, a contiguous stretch mapped to UniProt (Section 4), begins at an unobserved residue (dashed box), so its starting author number is missing and our code built the residue map from label numbers. It then looked atoms up by author number, which runs 34 ahead of label numbering, so the residue with author number 36 was paired with UniProt position 70 instead of 36. Under this shift the chain’s median IDDT was 0.198 at a median pLDDT of 98.4. With the mapping fixed it was 0.993.

et al., 2022, Li et al., 2026, Graber et al., 2025], and date-based or metadata-based splits leave similar overlap in docking [Buttenschoen et al., 2024] and protein–protein interaction benchmarks [Bushuiev et al., 2024, Kovtun et al., 2024]. Škrinjar et al. [2026] report that protein–ligand co-folding methods “largely memorize ligand poses from their training data”.

Exposure in AlphaFold evaluation. The AlphaFold2 paper evaluated the model on PDB chains released after its training cutoff and examined template similarity [Jumper et al., 2021]. BETA provides version-specific benchmark sets with sequence and structure filters [Dobson et al., 2025]. Terwilliger et al. [2024] compared AlphaFold predictions made without PDB templates with new crystal structures. AlphaFold’s predictions of fold-switched conformations have been traced to memorised training structures [Chakravarty et al., 2024], and retraining with parts of fold space withheld shows that generalisation survives large omissions but varies with what is withheld [Ahdritz et al., 2024].

Residue correspondence. The Worldwide PDB (wwPDB) remediated the archive to address inconsistencies introduced over the years [Henrick et al., 2008]. Mapping PDB chains to UniProt has long been non-trivial [Martin, 2005]. SIFTS provides residue-level mappings between the two [Velankar et al., 2013, Dana et al., 2019], records them in label numbering, and in author numbering where residues were observed, and documents the complications added by missing residues, insertions, expression tags and linkers [Choudhary et al., 2023]. Standard sequence alignments often misalign the regions next to unobserved residues [Velankar et al., 2013]. PDBrenum rennumbers PDB files to match UniProt [Faezov and Dunbrack, 2021]. Where these resources address how to map residues correctly, we document what one mapping mistake does to a downstream analysis.

Conformal prediction for structure. Split conformal prediction gives finite-sample marginal coverage under exchangeability [Vovk et al., 2005, Lei et al., 2018, Angelopoulos and Bates, 2023]. When proteins are sampled, residues pooled across proteins are generally not exchangeable, and such hierarchical data need their own guarantees [Dunn et al., 2023, Lee et al., 2026], as do non-exchangeable data more generally [Barber et al., 2023]. CalPro applies evidential and conformal methods to pLDDT [Shihab et al., 2026]. Our census registration includes a split-conformal coverage test (Section 7.5); we do not re-evaluate CalPro.

3 Exposure channels

Let R be an experimental reference structure, first released on date r , compared with an AFDB model M of the same protein. We distinguish three channels through which R , or structures like it, could have informed M .

Training. AlphaFold2 was trained on PDB structures with a maximum release date of 2018-04-30 [Jumper et al., 2021]. If r is on or before that date, R is *training-eligible*: it may have been a training example. Training also filtered chains and sampled them by cluster, so eligibility is an upper bound on use of the PDB itself. Self-distillation on predicted structures is a further route that release dates do not capture (Section 9).

Templates. At inference time, AlphaFold2 searches the PDB for templates. The AlphaFold human-proteome study reports the snapshots it used of the PDB and of PDB70, a clustered subset of the PDB used for template search [Tunyasuvunakool et al., 2021], and AFDB states that templates were restricted to structures released before 2021-02-15 [EMBL-EBI AlphaFold Database, 2026]. AFDB model files record the PDB entries supplied as templates in the ModelCIF category `_ma_template_ref_db_details` [EMBL-EBI AlphaFold Database, 2026, Vallat et al., 2023], so this channel can be measured for each model. The list records which templates the network was given, not how much the prediction relied on them.

Homology. Even when R post-dates both cutoffs, the same protein or a close relative may have been in the training-era PDB. Release dates cannot capture this channel; we measure it by sequence similarity to the training-era PDB [Jumper et al., 2021, Buttenschoen et al., 2024].

AFDB entries also carry a `modelCreatedDate`, documented as the date the entry was created [Google DeepMind, 2026]. In version 6, unchanged models received new metadata and the v6 label while their coordinates were carried over from version 4 [PDBe, 2025]. The creation date therefore records neither when a prediction ran nor what it could see. We use it only to exclude models created on or after the start of the preregistered census (Section 4).

4 Data and methods

Terms. We use *protein* for a UniProt accession, and call two structures *same-protein* if they map to the same accession. An *entry* is a PDB entry, which may contain several chains. A *chain segment* is the set of residues of one PDB chain mapped to one accession. SIFTS describes it as one or more *SIFTS segments*, each a contiguous stretch with its own start and end. A residue is a *confident error* if its pLDDT is at least 90 and its IDDT below 0.60. At the chain level we use a separate *low-agreement screen*, which flags a chain segment of at least 30 residues with median pLDDT ≥ 90 and median IDDT < 0.4 . The *training-era PDB* is the set of entries released on or before 2018-04-30, and an *accession-entry pair* is one protein in one PDB entry. A residue is *scored* if it has an IDDT value. Residue counts count each residue once per chain and entry, so a UniProt position observed in several chains is counted several times. A *registration* is an analysis plan, with its code and predictions, fixed in a version-controlled repository, now public (Data and code availability), and timestamped with OpenTimestamps [OpenTimestamps, 2026], which anchors a file’s SHA-256 hash in the Bitcoin blockchain. We call an analysis *preregistered* if it was registered before any IDDT value it uses existed.

ATLAS. The list of entries in the ATLAS download archive [DSIMB, 2026], retrieved on 2026-08-30, has 1,938 entry-chain records covering 1,735 PDB entries and 943 proteins, more than the datasets described in the ATLAS paper [Vander Meersche et al., 2024]. On 2026-09-04 we checked each record against PDBe and AFDB and kept chains that map to a single UniProt accession through SIFTS [Velankar et al., 2013, Dana et al., 2019], with the identity and coverage reported by PDBe at least 0.90 and 0.80, and whose protein has an AFDB model. Four of the 1,938 records could not be checked because of network errors. The kept chains cover 869 proteins. One has an AFDB record we could not parse, which leaves 868 proteins and 935 accession-entry pairs. For each reference we took the initial release date, the first record in the mmCIF category `_pdbx_audit_revision_history` [wwPDB, 2026].

Preregistered census. To obtain references released after both cutoffs of Section 3, we preregistered a census: every X-ray PDB entry at 2.5 Å resolution or better released after $T_0 = 2026-09-11\ 00:00\ \text{UTC}$, collected in weekly batches. Models come from AFDB version 6. A complete AFDB version 6 index was outside the registered access and rate-limit constraints, so the registered design builds its list of models incrementally: a protein enters the list when it first appears in the census, its model is accepted only if its

`modelCreatedDate` falls before the day of T_0 , and the model file is downloaded when its batch is processed. This is an operational rule based on metadata dates, which do not record when a prediction ran (Section 3). The design, analysis code and predictions were registered on 2026-09-09, and two dated amendments were made before any data were collected: one shortened the first batch to 48 hours and the other fixed the code that starts each batch. Apart from the release-date filter, which places every reference after AFDB’s 2021-02-15 template cutoff, the design adds no exposure control. What it adds is that every choice was fixed before any IDDT value was computed. The residue mapping was replaced later, after the error described in Section 7. We report the first four of 12 planned batches, covering releases up to 2026-10-04. The first, 48-hour batch contained no weekly PDB release day and is empty. With the original mapping the four batches gave 488 chain segments from 272 accession–entry pairs and 138 proteins, and with the corrected mapping 495 chain segments and 141 proteins; Table 2 in Section 7.2 shows how the batches narrow from enumerated entries to scored proteins. Most excluded chain mappings failed the registered AFDB version 6 admission checks (no eligible model found, metadata not retrievable, or a model file that was not version 6). The exposure measurements of Section 6 were made on the 138 proteins of the original set. The measurement for a given protein needs no residue pairing, but the error did decide which proteins were scored: the corrected mapping adds three (Section 7.2). All IDDT and coverage results use the corrected set unless stated otherwise.

Post-2022 census. An earlier census, assembled before the preregistered one and not preregistered, covers 168 X-ray entries at 2.5 Å or better released after 2022-06-02 and after the creation date recorded for their AFDB model. Its residue mapping used the same numbering fallback but discarded pairs whose amino acids differed, and it was affected by the same error. It scored chain A of each entry only. Its candidates were the X-ray entries at 2.5 Å or better released after 2022-06-02 (33,517 in an RCSB search on 2026-09-05). The step that reduced them to 168, which included a filter on entry size (fewer than 5,000 atoms), is not part of the released code; only its resulting list is released. We use it for the exposure comparison of Section 6.3 and to check how the error behaves under a per-pair identity filter (Section 7.4). With the corrected mapping, 139 of its entries from 85 proteins are scored (96 entries from 64 proteins before). All its results outside Section 7.4 use the corrected IDDT values.

Model versions. The cutoffs of Section 3 are those of AlphaFold2, but AFDB also serves ColabFold models [Mirdita et al., 2022], so we checked the software recorded in each model file. All 138 model files of the preregistered census’s original exposure roster are AlphaFold2 models (“AlphaFold Monomer v2.0”). Of the 93 model files downloaded for the post-2022 census, 6 are ColabFold Monomer v1.5.2 models and 87 are AlphaFold Monomer v2.0, and none of the six belongs to a protein in the exposure comparison of Section 6.3.

Residue pairing and IDDT. For each reference chain we pair $C\alpha$ atoms with AFDB model residues through the PDBe SIFTS UniProt mapping. We describe the corrected mapping here, and Section 7 the original and its error. The corrected code (i) reads both the SIFTS positions and the atoms in label numbering (`label_seq_id`), with chains identified by their label identifier (`label_asym_id`, called `struct_asym_id` in the SIFTS files), so that no lookup crosses numbering schemes; (ii) maps a SIFTS segment whose PDB and UniProt ranges have equal length by a constant offset, drops any pair whose amino acids differ, and rejects the segment if more than 10% of its pairs differ; and (iii) otherwise aligns the deposited sequence within the segment’s label range to the AFDB model sequence within its UniProt range (global alignment; match +2, mismatch −1, gap −2; ties favour pairing over a gap) and keeps only identical aligned pairs, so the 10% rule cannot reject these segments. Segments with fewer than 30 paired residues are excluded, and IDDT is computed per SIFTS segment; in the four batches every scored chain segment consists of a single SIFTS segment, so this is the same as computing it per chain segment. We use $C\alpha$ atoms that have no alternative location or are at location A, and exclude modified residues other than selenomethionine; in the four batches with the corrected mapping these two rules remove 16 and 57 residues. We score each residue by IDDT- $C\alpha$ [Mariani et al., 2013], computed against the AFDB model over paired residues with a 15 Å inclusion radius and distance-difference thresholds of 0.5, 1, 2 and 4 Å. pLDDT is read from the model’s B-factor column and is never used in computing IDDT.

Homology exposure. For each census protein we queried the search service of the RCSB PDB (Research Collaboratory for Structural Bioinformatics; Burley et al., 2023) for training-era entries in three classes: (i) entries mapped to the same UniProt accession, (ii) entries containing a chain of a different accession at $\geq 95\%$ sequence identity and (iii) entries containing one at $\geq 30\%$ identity. Classes (ii) and (iii) come from MMseqs2 sequence searches [Steinegger and Söding, 2017] with an E-value of at most 0.1 and the canonical UniProt sequence as query; identity is measured over the aligned region the search returns. We set no coverage threshold, so a hit of any class, even a same-protein hit, may cover only part of the protein, and the absence of a hit does not rule out remote homology. Each protein is assigned its strongest class. A failed query would leave a protein unresolved rather than novel, but none failed: for the 224 proteins queried (138 from the preregistered census and 86 from the post-2022 census, namely the proteins scored there before or after correction; one of them, Q2G0L4, is not among the 85 scored after correction because the identity checks reject both its entries), 594 queries returned hits, two of them repeats of an identical query, and 408 returned none [RCSB Protein Data Bank, 2026b].

Template exposure. From each AFDB model file we read the template PDB IDs, mapped each template to UniProt through SIFTS and recorded its release date. A model has a *same-protein template* if any template entry contains a chain that maps to its accession. For 44 of the 45 such models in the preregistered census, the template chain named in the model file itself maps to the accession. For the remaining one (Q5UE59, template 6SWU), the model file names the chains AAA and FFF while SIFTS names them A to F; every chain of 6SWU maps to Q5UE59. It is *unresolved* if none does and at least one template could not be mapped.

Preregistered coverage prediction. The census registration also predicts the coverage of split-conformal intervals for residue IDDT that are built without pLDDT, so the prediction is sensitive to the IDDT values and the pipeline rather than to pLDDT; we call its evaluation the coverage test. Section 7.5 describes it with its result.

5 Most ATLAS references predate the training cutoff

For ATLAS we measured only the training channel. We count a protein as training-eligible when all of its retained references were released on or before 2018-04-30. Of 868 ATLAS proteins, 791 (91.1%) are training-eligible in this sense, and so are 851 of the 935 accession–entry pairs (91.0%). The references were released between 1988 and 2023 (median release year 2008 over the 935 pairs). Only 77 proteins have any reference released after the cutoff, so agreement between AFDB models and ATLAS references cannot by itself be read as evidence of generalisation. Comparing each reference’s release date with its model’s `modelCreatedDate` instead, a tempting but wrong proxy (Section 3), would make all references of 863 of 868 proteins (99.4%) predate their model.

ATLAS was built to describe dynamics, and reusing it to study AlphaFold error requires an explicit exposure control. The obvious control is to keep only references released after the cutoff. Section 6 asks, on a separate set of newly released structures, how much that achieves.

6 A post-cutoff release does not imply an unseen protein

This section measures the homology and template channels in the preregistered census, whose references were all released after both cutoffs (Sections 6.1 and 6.2). It then asks, in the post-2022 census, whether exposure goes with higher agreement (Section 6.3).

6.1 Homology exposure

Every structure in the preregistered census was released more than eight years after the training cutoff, yet 87 of the 138 proteins (63%) had a same-protein or near-identical ($\geq 95\%$ identity) structure in the training-era PDB, 128 (93%) had a relative at $\geq 30\%$ identity, and only 10 (7%) had none (Table 1). For same-protein hits, the earlier structure may cover only part of the protein. Some well-studied proteins have hundreds of

Table 1: Of the 138 proteins in the first four batches of the preregistered census, 63% had a same-protein or $\geq 95\%$ -identical structure in the training-era PDB and 33% were given a same-protein template. Homology: class of the strongest relative, one per protein. Templates: as listed in the AFDB model file. The protein set is that of the original residue mapping; the three proteins added by the corrected mapping (Section 7.2) are not included.

Class	Proteins	Share
<i>Homology (training-era PDB)</i>		
Same protein (UniProt accession)	78	56.5%
Different accession, $\geq 95\%$ identity	9	6.5%
Different accession, $\geq 30\%$ and $< 95\%$ identity	41	29.7%
No relative found at $\geq 30\%$ identity	10	7.2%
<i>Templates (AFDB model file)</i>		
Same-protein template	45	32.6%
Other templates only	88	63.8%
Unresolved ^a	5	3.6%

^aAt least one template could not be mapped to UniProt through SIFTS and none of the others maps to the protein, so whether the model had a same-protein template is unknown.

such structures: hen egg-white lysozyme (P00698) had 730 same-protein entries in the training-era PDB, human carbonic anhydrase 2 (P00918) had 700 and β_2 -microglobulin (P61769) had 693. These hits show only that a related structure existed, not how it differs, for example by ligand, mutation, construct or crystal form.

6.2 Template exposure

The template lists give a more direct measure. All 138 model files list four PDB templates each, released between 1988 and January 2021, so none post-dates the documented 2021-02-15 cutoff. For 45 of the 138 models (33%) at least one template is a structure of the same protein, for 88 all templates are of other proteins, and 5 are unresolved (Table 1). The homology and template channels overlap only in part. Of the 78 proteins with a same-protein structure in the training-era PDB, 43 were given a same-protein template and 35 were not; none of the 35 is unresolved. Conversely, two proteins with no same-protein structure in the training-era PDB were given a same-protein template (released in 2019 and 2020). One of them, Q9F0J8, has no training-era relative at $\geq 30\%$ identity, so a homology screen alone would call it novel although its model was given a structure of the same protein as a template.

6.3 Is exposure associated with agreement?

This comparison uses the post-2022 census, for a reason given at the end of this section. We compared its proteins with and without a *close relative* in the training-era PDB (a same-protein structure or one at $\geq 95\%$ identity), using IDDT values recomputed with the corrected mapping (Section 7.4). We fixed the comparison in a timestamped plan before running it, but after its IDDT values existed, so it is exploratory. For each protein we took the median IDDT and the median pLDDT over its residues, and for each group we then averaged these per-protein medians. Intervals are 95% percentile bootstrap intervals from 10,000 resamples of proteins within each group. Of the 85 scored proteins, the 40 with a close relative had a mean median IDDT of 0.975, against 0.959 for the 45 without one, of which 38 have a relative at at least 30% but below 95% identity and 7 have none found (difference 0.016, interval 0.003–0.032). Their mean median pLDDT was also higher, 96.1 against 92.3 (difference 3.8, interval 1.7–6.0), a larger gap than in IDDT (3.8 against 1.6 points on a 0–100 scale), although we did not test the difference between the two gaps formally. The mean per-protein share of confident errors, counted over all scored residues of each protein, was 0.12% with a close relative and 0.13% without (difference -0.01 percentage points, interval -0.18 to 0.17). The interval is wider than either share, so the data cannot tell the groups apart on this measure. These are associations in a small sample: proteins with close relatives may also differ in size, family and how often they have been

studied. We did not run this comparison on the preregistered census because its registration also makes predictions about coverage within pLDDT bins that are not yet evaluated, and comparing pLDDT with IDDT there first would compromise their blindness.

7 A residue-correspondence error created spurious confident errors

The exposure measurements need no residue pairing, but any comparison of agreement does. It needs per-residue IDDT values, and computing them for the census is where our own pipeline failed. After the fourth batch the preregistered coverage prediction, which does not use pLDDT, failed at both nominal levels, 0.90 and 0.95 (Section 7.5). A post hoc check found that one protein, K7PQ54, held 14% of the scored residues and about half of the residues that fell outside the 90% intervals, counted as test-set occurrences pooled over the 200 splits. Tracing it led to the numbering fallback, an error that makes the model look worse than it is, whereas exposure can make it look better. The registration and the stored source files let us confirm the cause and rescore the census.

7.1 Mechanism

In the segment-level PDBe SIFTS mapping files we used (PDBe also distributes per-residue mappings), each SIFTS segment has a start and an end in label numbering and, when the boundary residue was observed, in author numbering. Our census code built each SIFTS segment’s residue map from its starting author number and fell back to the starting label number when the author number was missing, which happens whenever the first residue of the segment is unobserved. Our code then looked atoms up by author number. In a segment where the fallback was used and author numbers differed from label numbers by a constant offset, every residue was paired with a UniProt position displaced by that offset, a *register shift*. The code had no amino-acid identity check to catch it.

Figure 1 shows chain A of entry 9qj6, a structure of K7PQ54, dihydrolipoyl dehydrogenase from the lesser grain borer *Rhyzopertha dominica* (507 residues; The UniProt Consortium, 2025). The SIFTS segment starts at label residue 1 (UniProt 35), which is unobserved. Author numbering runs 34 ahead of label numbering, so the atom with author number 36 (label 2, UniProt 36) was assigned UniProt 70. Under this shift chain A had a median IDDT of 0.198 at a median pLDDT of 98.4, and chain B, shifted in the same way, had 0.199. With the mapping fixed, chains A and B paired 472 and 471 residues with no sequence mismatch and had median IDDT 0.993 and 0.995.

7.2 How many chain segments the error affected

We rescored the four batches from their stored reference mmCIF, SIFTS and AFDB model files, after checking their hashes, once with each mapping. With the original mapping the rescoring reproduced the stored IDDT values of every batch exactly, which confirms that it re-executes the preregistered pipeline. One entry that meets the census criteria, 9zxa (protein Q92802, 8 chains), is absent from both sets because the original run never downloaded its reference file. Table 2 compares the two sets of IDDT values.

Correcting the mapping changed the median IDDT of 188 of the 488 original chain segments. For 151 of them, from 40 proteins, the change exceeded 0.2, and all 151 moved upward (Figure 2). The other 37 changed by at most 0.056. After correction the low-agreement screen flags no chain segment, and 13 entries in which two chains had differed by 0.4 or more in median IDDT no longer differ that much.

The corrected set is not a superset of the original. It scores 6,465 residue positions that the original did not and drops 916 that it did, a net gain of 5,549, and it adds three proteins and seven chain segments that the original mapping had rejected. A reimplementaion with a separately written parser and IDDT code reproduced the stored corrected IDDT values (four decimals) of 36 chain segments exactly: 26 chosen from proteins affected by the error or with unusual chain identifiers, and 10 unchanged segments drawn at random. A second check, with a different mmCIF parser and IDDT implementation [Kunzmann and Hamacher, 2018] and PDBe’s per-residue SIFTS mapping, which gives each residue’s UniProt position directly rather than through segment boundaries, reproduced 494 of the 495 corrected chain segments, every IDDT value within

Table 2: The correction clears every low-agreement flag. First four batches of the preregistered census, from enumerated entries to scored residues, with the original and the corrected residue mapping.

	Original	Corrected
PDB entries enumerated	409	409
Entries with a parsed SIFTS mapping ^b	364	365
Entries with a scored chain segment	218	220
Accession–entry pairs	272	275
Chain segments	488	495
Proteins	138	141
Scored residues	117,905	123,454
Residues with IDDT < 0.60	31.9%	1.1%
Chain segments flagged by the low-agreement screen ^a	116	0
K7PQ54 chain segments with median IDDT < 0.4	29 of 38	0 of 38

^aAt least 30 residues, median pLDDT ≥ 90 and median IDDT < 0.4. The 116 come from 32 proteins. ^bThe corrected parser also reads the mapping of entry 9zxa, which neither run could score because its reference file was never downloaded.

5×10^{-5} . In the remaining one, chain A of entry 21xg, two crosslinked residues leave the global alignment of step (iii) with two equally good solutions, and ours pairs one residue with the UniProt position next to the one that the per-residue mapping gives, so each mapping has one position the other lacks. On the shared positions this moves 18 IDDT values by up to 0.054 and the chain’s median by 0.003, and leaves the share of residues below 0.60 unchanged.

7.3 The error mimics overconfidence, and a SIFTS-segment identity check catches it

The error created residues with high pLDDT and very low IDDT, the pattern an overconfident model would produce. A calibration study run on these values could have concluded that AlphaFold2 is overconfident on new structures, and our preregistered coverage test, which does not use pLDDT, also failed on them (Section 7.5).

The fallback is triggered by a SIFTS segment that begins on an unobserved residue, and this is common: 304 of the 492 SIFTS segments mapped to the 138 original proteins (62%) begin this way. It causes a shift when author and label numbering are offset at the start of the segment, and an offset that changes partway along a segment causes further shifts. A nonzero offset at the segment start occurred in 147 of the 492 SIFTS segments (30%), from 39 proteins, with offsets (author minus label number) from -31 to $+201$. In a post hoc comparison, all 147 lay within the 151 chain segments whose median IDDT changed by more than 0.2. Each of the other four of those 151 contains a SIFTS segment whose offset changes partway along it.

Checking amino-acid identity for every pair is necessary but not sufficient. When we applied 18 register shifts to four corrected chains, only 5–7% of pairs matched by chance, and such pairs pass a per-pair filter. What catches the shift is the per-pair check together with rejecting any SIFTS segment in which more than 10% of pairs differ. These checks are diagnostics, not guarantees: a pair can pass them and still sit one position off in an ambiguous alignment, as in chain A of 21xg (Section 7.2). The post-2022 census had only the per-pair check. It therefore lost most shifted residues but kept those that matched by chance (Section 7.4). Coarser checks raise alerts, though real conformational differences can also trigger them. A set in which 31.9% of residues fall below 0.60 IDDT is far from the 2.6% in a separate set, the corrected post-2022 census, and 13 entries had chains that differed by 0.4 or more in median IDDT. A chain-consistency check, which compares median IDDT across an entry’s chains, would still miss entries such as 9qj6, where both chains were shifted alike.

7.4 The same error in the post-2022 census

Of the 168 entries of the post-2022 census, 96 had originally been scored, 69 had been rejected for having too few paired residues and 3 because the model could not be downloaded. With the corrected mapping, 45 of

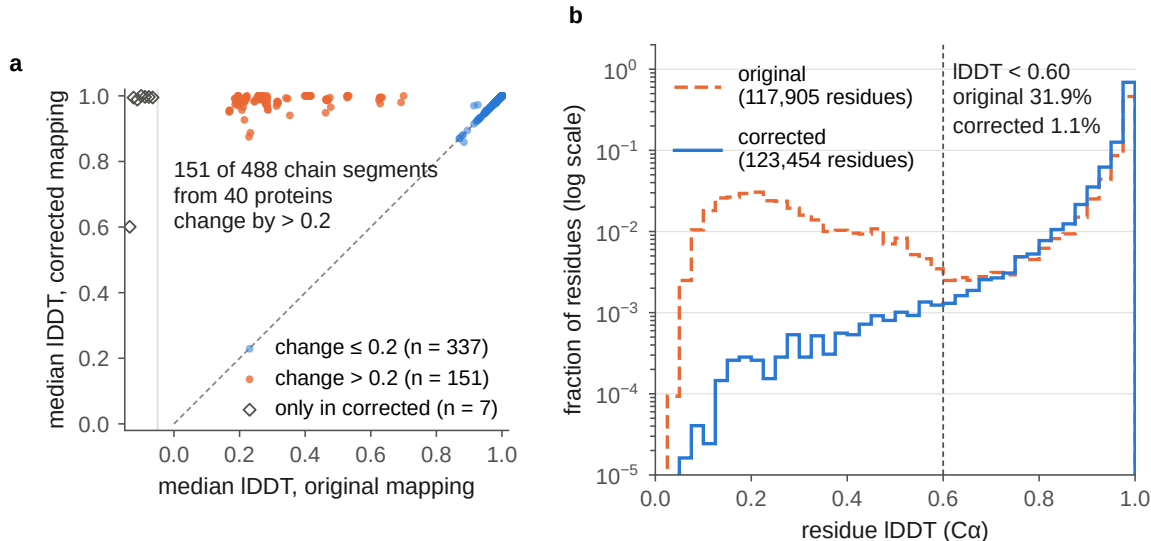


Figure 2: The correction raised the median IDDT of every chain segment that it changed by more than 0.2. (a) Median IDDT per chain segment with the original (x) and the corrected (y) mapping. The change is the absolute difference between the two medians, and the dashed line marks no change. The 151 chain segments with a change above 0.2 (orange) moved from original medians of 0.17–0.70 to corrected medians of at least 0.87. Diamonds in the strip left of the axis, spread sideways for visibility, are the seven chain segments that only the corrected mapping scores; they have no original value. (b) Distribution of residue IDDT under each mapping: fraction of each mapping’s scored residues per bin of width 0.025 (logarithmic y axis). The two sets of residues are not identical.

those 69 entries yield IDDT values. In all, 139 entries are scored instead of 96 (37,531 scored residues instead of 24,457). The total is $96 + 45 - 2$ because two previously scored entries, 10pa and 10ps, are now rejected by the identity checks. The Spearman rank correlation between pLDDT and IDDT rose from 0.617 to 0.643. The pooled residue-level share of confident errors, taken over all residues with pLDDT ≥ 90 , halved from 0.58% (121 of 20,777) to 0.29% (90 of 31,032); unlike the per-protein shares of Section 6.3, its denominator is residues with pLDDT ≥ 90 . In the lowest-confidence bin (pLDDT < 50) the share below 0.60 IDDT rose from 58% ($n = 177$) to 75% ($n = 314$).

The added residues and the rise in the lowest-confidence bin come from the 45 newly scored entries. Most of the rise in rank correlation, and more than half of the fall in confident errors, come from recomputing entries that were already scored, where shifted residues had matched by chance and passed the per-pair filter. Entry 10ke, for example, held 22 of the 121 original confident errors. The input files of the original post-2022 run, downloaded in September 2026, had not been kept, so this recomputation used files downloaded again in October 2026; changes to those files between the two downloads could account for part of the difference.

7.5 The preregistered coverage prediction before and after correction

The census registration includes a prediction that is sensitive to the pipeline rather than to AlphaFold2’s confidence. Let y be a residue’s IDDT and $1 - \alpha$ the nominal coverage. For each of 200 random protein-level splits (seeds 0–199), 50% of proteins form a fitting set, 25% a calibration set and 25% a test set. The fitting set serves only to compute $\hat{y}$, a single number: the median IDDT over its residues. Every test residue receives the interval $\hat{y} \pm q$, where q is the $[(n_{\text{cal}} + 1)(1 - \alpha)]$ -th smallest of the calibration scores $|y - \hat{y}|$ and n_{cal} is the number of calibration residues [Vovk et al., 2005, Lei et al., 2018]. At $1 - \alpha = 0.90$ and 0.95 , we predicted that mean residue-level test coverage over the 200 splits would be at least the nominal level minus three Monte Carlo standard errors (MCSE, the standard deviation of coverage across splits divided by $\sqrt{200}$), a threshold that depends on the data. Residues enter the test if their stored IDDT, pLDDT, Kabsch distance and B-factor z-score are finite and $0 \leq \text{pLDDT} < 100.01$; pLDDT is not an input to the interval or the nonconformity score. The MCSE measures variation over random splits of these proteins, not

Table 3: The preregistered coverage prediction fails on the original IDDT values and meets its registered rule on the corrected ones. Mean: residue-level test coverage averaged over 200 protein-level splits. Threshold: nominal level minus three MCSE, where the MCSE (Monte Carlo standard error) is the standard deviation of coverage over the splits divided by $\sqrt{200}$. Margin: mean minus threshold.

Nominal	Original IDDT values			Corrected IDDT values		
	Mean	Threshold	Margin	Mean	Threshold	Margin
0.90	0.8580	0.8760	-0.0180	0.8945	0.8914	+0.0031
0.95	0.9152	0.9328	-0.0176	0.9456	0.9449	+0.0007

uncertainty about new proteins. Calibration residues are pooled across proteins of different sizes, and for such hierarchical data marginal residue-level coverage carries no finite-sample guarantee [Dunn et al., 2023, Lee et al., 2026].

On the original IDDT values the prediction failed at both levels (Table 3). Because those values were corrupted by the numbering fallback, we withdraw this outcome as invalid; a dated, timestamped correction notice in the repository states this and why (Appendix A). The original outputs remain archived unchanged. On the corrected values (123,454 residues from 141 proteins) both means are slightly below nominal but pass their thresholds. Meeting this rule shows neither that coverage is nominal nor that the IDDT values are right: values all rescaled by the same positive factor and shift would give the same coverage in exact arithmetic. At 0.90 the mean exceeds the threshold by 0.0031, about one MCSE. At 0.95 it exceeds it by 0.0007, less than half an MCSE (0.0017), so this level passes narrowly and could reverse under a different but equally defensible correction of the residue mapping. IDDT takes a limited set of values and is stored to four decimals, so some test residues lie exactly on an interval end $\hat{y} \pm q$; counting them as uncovered leaves the pass intact (0.9453 against 0.9449). The original thresholds were lower because the original MCSE (0.0080 at 0.90, 0.0057 at 0.95) was more than twice the corrected one: K7PQ54 alone held 14% of the original residues, so a split’s coverage depended strongly on whether K7PQ54 fell in its test set.

The seeds and the decision rule were fixed by the original registration, and the corrected mapping and the protocol for rerunning the test on corrected values were registered before the rerun. The rerun is not blind. Before registering it, we computed the same statistic outside the registered pipeline on an early version of the corrected mapping (122,024 residues rather than 123,454) and obtained 0.898 and 0.947. The rerun used the registered evaluation code unchanged, and no new batches were collected until it finished. Two reimplementations written from the registration, without importing the evaluation code, reproduced both means exactly, and one also reproduced the original failure. The corrected pass, together with the mechanism in Section 7.1, makes the original failure consistent with an artefact of the residue mapping. Because the interval ignores pLDDT, the result says nothing about how well pLDDT tracks error; what it showed is that the test was sensitive to this mapping error in this census.

8 Recommendations

Table 4 lists checks for the two problems reported here, with the evidence for each. We recommend the following.

1. Report exposure by channel, and state which channels the question requires to be controlled. Release dates alone measure only training eligibility, and AFDB creation dates measure none of the channels (Section 5).
2. For AFDB models, read the template list from the model file (Section 6).
3. Stratify results by sequence identity to the training-era PDB [Buttenschoen et al., 2024], or split data by similarity [Joeres et al., 2025] (Section 6).
4. Pair residues in one numbering scheme, preferably from per-residue mappings where available, check amino-acid identity for every pair, reject any SIFTS segment with more than 10% mismatches, and run the coarser checks of Table 4 before any downstream analysis (Section 7.3).

Table 4: Checks for AFDB–experiment comparisons and the evidence for each in this paper. Thresholds are operational choices, not validated optima, and the last four checks are diagnostics rather than guarantees.

Check	What it catches	How	Evidence here
Training eligibility	Possible training example	Release date on or before 2018-04-30	ATLAS: 791 of 868 proteins training-eligible (Section 5)
Template exposure	Same protein as template	Template list in the model file, mapped through SIFTS	Preregistered census: 45 of 138 models (Section 6)
Homology exposure	Close training-era relative	Sequence search of the training-era PDB	Preregistered census: 87 of 138 proteins with a same-protein or $\geq 95\%$ -identical relative; 10 with none at $\geq 30\%$ (Section 6)
Identity check	Register shifts	Amino acids of each pair match; reject offset-mapped SIFTS segments with $>10\%$ mismatches	Under a register shift only about 5–7% of pairs match by chance, so the segment rule rejects shifted segments; the per-pair check alone kept chance matches in the post-2022 census (Sections 7.3, 7.4)
IDDT distribution	Gross corruption	Compare the share of IDDT < 0.60 with a separate set	31.9% original, against 2.6% in the corrected post-2022 census (Section 7.3)
Chain consistency	Shifts in one chain	Compare median IDDT across an entry’s chains (flag differences ≥ 0.4)	13 entries flagged, 9qj6 missed (Section 7.3)
Registration, stored inputs	Lets errors found later be diagnosed and corrected	Fix the design in advance, keep hashed inputs	Rescoring with the original mapping reproduced the stored IDDT values of every batch (Section 7.2)

- Before attributing a failed calibration or coverage test to the model, rule out errors in the reference values. Our failed coverage test, which does not use pLDDT, is consistent with an error in the IDDT values (Section 7.5).
- Register analyses before IDDT values exist and keep the raw inputs (Sections 7.2 and 7.4).

9 Limitations

Our training channel uses AlphaFold2’s 2018-04-30 cutoff. Other versions, including AlphaFold3 [Abramson et al., 2024], have their own cutoffs, and our shares do not transfer to them. We did not measure exposure through multiple sequence alignments or self-distillation, and template lists do not show how much a prediction relied on its templates (Section 3). Homology was measured with fixed identity thresholds, a single canonical sequence and no coverage threshold. A $\geq 30\%$ hit can be a short domain match, which overstates homology exposure, and a single canonical sequence can miss isoform-specific hits, which understates it. IDDT compares one experimental chain with a monomer model, so partner chains, ligands or crystal packing can change the experimental conformation and lower IDDT for reasons unrelated to exposure. We score $C\alpha$ IDDT, so our values are not directly comparable with all-atom IDDT.

The preregistered census samples new PDB entries, which over-represent proteins that are crystallised repeatedly, so its exposure shares describe what a post-cutoff release filter yields rather than the novelty of proteins in general. It is reported after four of its 12 batches, and the full census may differ. Because the correction adds and drops residues, the before-and-after coverage comparison is not made on identical residues (Section 7.2), and because the correction changes which proteins are scored, the same seeds do not give the same splits, so the two results are not a paired comparison. The corrected coverage result comes from a rerun registered after a preliminary result was known, and it passes at 0.95 by less than half an MCSE (Section 7.5). The numbering fallback is documented in one pipeline, ours. The trigger rates we report (Section 7.3) describe this census, and we have not examined whether any published analysis is affected. The post-2022 census scored chain A only, and the step that selected its 168 entries is not released.

The census model list rests on metadata dates (Section 4). The bootstrap intervals of Section 6.3 resample proteins and ignore relatedness between them. Finally, the three exposure channels overlap and do not cover every route (Section 3).

10 Conclusion

Agreement between an AFDB model and an experimental structure is evidence about AlphaFold’s generalisation to unexposed proteins only if two things are known: what the model could have seen, and that the residues being compared are the same residues. In ATLAS, 91% of proteins had only training-eligible references. In the preregistered census, whose references all post-date the training cutoff, 63% of proteins had a same-protein or $\geq 95\%$ -identical structure in the training-era PDB. In an exploratory comparison on an earlier census, proteins with such a relative had slightly higher IDDT and higher pLDDT. In our own pipeline a numbering fallback, with no SIFTS-segment identity check to catch it, made accurate predictions look like confident errors. A preregistered coverage prediction, which does not use pLDDT, failed on the resulting values; a rerun registered after the error and a preliminary result were known meets it on corrected values, narrowly at the 0.95 level. Registration, stored inputs and the checks in Table 4 are cheap enough to be routine for AFDB comparisons.

A Provenance

This appendix records when each step of the preregistered census and its correction happened, so that readers can check which decisions preceded which data (Table 5). The README of the repository (Data and code availability) maps each section, table and figure of this paper to its registration, code and output files. The correction notice is a dated document, timestamped like the registrations, that withdraws the original outcome of the coverage test and states why. Registrations and intermediate outputs carry timestamped hashes. The repository also holds the reports of the independent checks of Sections 7.2 and 7.5, each made with separately written code.

Data and code availability

The registrations with their timestamp proofs, the analysis code, the derived residue-level tables, the SHA-256 hashes of the downloaded files and web responses, and the reports of the independent checks are available at <https://github.com/MachineLearning-Nerd/afdb-exposure-audit>. A script in the repository runs 220 checks: 157 recompute numbers in this paper from the row-level tables and cached responses, 43 check stored outputs against those recomputations, and 20 compare numbers in the paper with stored outputs. It checks numbers, not the prose. The census’s other registered predictions have not been evaluated; its evaluation outputs, which contain fields for them, are withheld until the census closes. Raw files from the PDB, AFDB, PDBe/SIFTS, UniProt and ATLAS are not redistributed but can be re-fetched, and the recorded hashes identify the exact files used. The ATLAS counts in Sections 4 and 5 are computed from released tables. One intermediate step, which sorted the 869 proteins into two lists for a separate, unpublished study (this paper uses both lists), reads a file that is not released, so that step cannot be rerun; its outputs are included. Likewise, the step that selected the 168 entries of the post-2022 census is not released; the resulting list is.

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Table 5: Timeline of the preregistered census and its correction. Events on the same date are listed in the order in which they occurred, except the exposure-analysis rows, which are listed last for their date.

Date (2026)	Event
9 Sep	Design, analysis code and predictions registered (OpenTimestamps proof confirmed in Bitcoin on 10 Sep)
10 Sep	Two dated amendments made before any data were collected: batch 1 shortened to 48 h; fix to the code that starts each batch
11 Sep	Census start T_0 : only entries released later, with AFDB version 6 models created before this date, qualify
4–5 Oct	Evaluation after batch 4: coverage prediction fails at both levels
7 Oct	Post hoc check traces the low IDDT of K7PQ54 (entry 9qj6) to the numbering fallback; data collection paused
7 Oct	Unregistered preliminary calculation on an early version of the corrected mapping (mean coverage 0.898 at 0.90, 0.947 at 0.95)
7 Oct	Corrected mapping, protocol for rerunning the coverage test, and correction notice registered
7 Oct	Batches 1–4 rescored from stored files with both mappings; post-2022 census rescored with the corrected mapping from re-downloaded files
7 Oct	Independent check of the corrected mapping, the rescoring and the post-2022 recomputation
7 Oct	Plan for the training and homology exposure analyses (Sections 5, 6.1) written; homology searches run
8 Oct	Addenda for the extended homology, template, association and trigger analyses (Sections 6.2, 6.3, 7.3) written and timestamped; analyses run
9 Oct	Second label check with an independent parser over all 495 chain segments (Section 7.2)
9 Oct	Registered rerun of the coverage test on corrected IDDT values completes; meets the rule at both levels
9 Oct	Two independent reimplementations reproduce the rerun

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